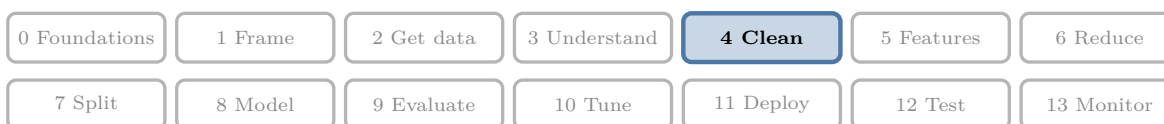


Outlier Detection with the IQR Method

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Where this fits: Pipeline map step 4 (Clean). See the [Course map](#).



- **Builds on:** Descriptive statistics ([Note 19](#)); Univariate analysis ([Note 20](#)); Outliers ([Note 41](#)).
- **Compare with:** Z-score outlier method ([Note 42](#)).

1. Overview

Key point: For a skewed column, every value beyond the box-plot fences ($Q_1 - 1.5 \text{ IQR}$ and $Q_3 + 1.5 \text{ IQR}$) is an outlier; we then trim those rows or cap the values at the fences.

Note 41 listed three rules for detecting outliers, and Note 42 put the first one, the z-score method, to work on a normal column. This Note covers the second rule: the **IQR method**, also called the **IQR proximity rule**. It is the rule to use when a column is skewed.

Figure 1 shows the whole method. We check that the column is skewed, compute the two fences from the quartiles, and then treat the values outside them by trimming or capping (both defined in Note 41, Section 7).

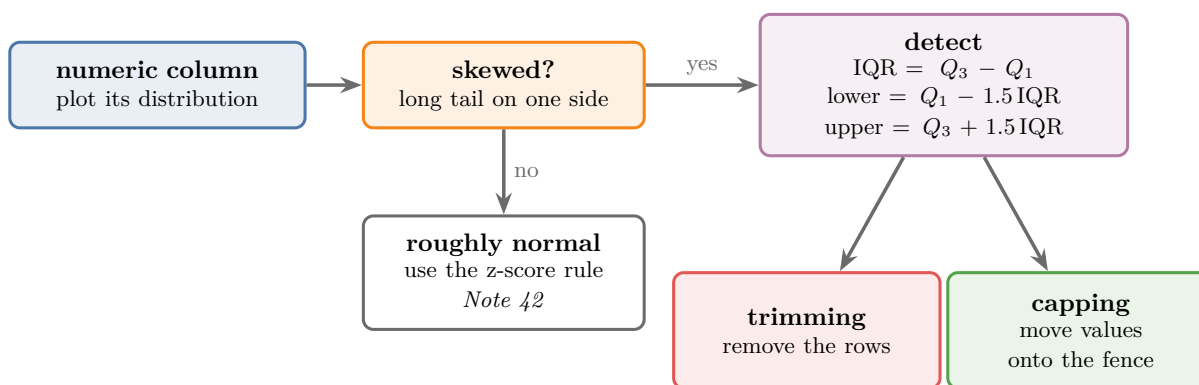


Figure 1: The IQR method: check the column is skewed, compute the fences, then trim or cap

2. The condition: a skewed column

Key point: The IQR method is meant for a column that is skewed, not normal.

A column is **skewed** when its values have a long tail on one side (Note 20). The z-score method of Note 42 does not fit such a column, because it assumes a bell shape.

The IQR method makes no such assumption. It is built only on percentiles, which do not care about the shape of the column. To use it we need two ideas from Note 20: the box plot

and the IQR.

3. Percentiles, quartiles and the IQR

Key point: The p -th percentile is the value below which $p\%$ of the column lies; Q_1 , the median and Q_3 are the 25th, 50th and 75th percentiles, and the IQR is $Q_3 - Q_1$.

A **percentile** splits a column by rank. The p -th percentile is the value that $p\%$ of the values lie below.

- **100th percentile:** every value is at or below it, so it is the maximum.
- **0th percentile:** no value is below it, so it is the minimum.
- **25th percentile (Q_1):** a quarter of the values lie below it.
- **50th percentile:** half the values lie below it, so it is the **median**.
- **75th percentile (Q_3):** three quarters of the values lie below it.

Q_1 and Q_3 are the **quartiles**: they cut the sorted column into quarters. A box plot draws its box from Q_1 to Q_3 , with a line at the median (Note 20, Figure 7).

The **interquartile range (IQR)** is the width of that box, the range that holds the middle half of the data:

$$\text{IQR} = Q_3 - Q_1$$

4. The fences

Key point: The lower fence is $Q_1 - 1.5 \times \text{IQR}$ and the upper fence is $Q_3 + 1.5 \times \text{IQR}$; values outside them are outliers.

A box plot does not draw its whiskers to the smallest and largest values. It stops them at two limits, the **fences**, set 1.5 IQR beyond the edges of the box (Note 20, Section 8.1). Every value outside the fences is drawn as a dot: an outlier.

The IQR method uses exactly these fences as its lower and upper limits:

1. **In words:** take the width of the box, multiply it by 1.5, and step that far out from each edge of the box.
2. **Formula:**
$$\text{lower} = Q_1 - 1.5 \times \text{IQR}, \quad \text{upper} = Q_3 + 1.5 \times \text{IQR}$$
3. **Example:** the placement exam marks of Section 6 have $Q_1 = 17$ and $Q_3 = 44$, so $\text{IQR} = 44 - 17 = 27$ and $1.5 \times 27 = 40.5$:

$$\text{lower} = 17 - 40.5 = -23.5, \quad \text{upper} = 44 + 40.5 = 84.5$$

A mark below -23.5 or above 84.5 is an outlier.

The plan for any skewed column is therefore short: compute Q_1 , Q_3 and the IQR, compute the two fences, and trim or cap every value outside them.

Extra: The fences are robust. Q_1 and Q_3 depend only on the order of the middle values, so a few extreme values cannot drag them out, unlike the mean and standard deviation of Note 42. The factor 1.5 is a convention from John Tukey, who invented the box plot; some people use 3 to flag only “extreme” outliers.

5. Treating the outliers: trimming or capping

Key point: Trimming deletes the outlier rows; capping replaces each outlier with the fence it crossed.

The two treatments are the same as in Note 42:

- **Trimming:** remove every row whose value lies outside the fences.
- **Capping:** keep every row, but move each value above the upper fence down to the upper fence, and each value below the lower fence up to the lower fence.

Trimming is simplest. When many rows are outliers, capping keeps the data from shrinking.

6. The placement data

Key point: The data has 1,000 students; their placement exam marks are right-skewed (skewness 0.84), so the marks column is the one for the IQR method.

The data is the placement data of Note 42: one row per student of a college, 1,000 students, three columns.

cgpa	placement_exam_marks	placed
7.19	26	1
7.46	38	1
7.54	40	1
6.42	8	1

- **cgpa:** the student’s CGPA (out of 10).
- **placement_exam_marks:** marks out of 100 in the aptitude test held before placement.
- **placed:** 1 if the student got a job offer, 0 if not.

Note 42 (Figure 3) showed the shapes: **cgpa** is a bell (skewness -0.01), while **placement_exam_marks** has a long tail to the right (skewness 0.84). So **placement_exam_marks** is the candidate for the IQR method.

The summary numbers of the marks column tell the same story:

Count	Mean	Std	Min	25%	50%	75%	Max
1,000	32.23	19.13	0	17	28	44	100

A quarter of the students scored below 17, half below 28 and three quarters below 44. One student scored 0 and one scored the full 100, which suggests a hard exam with a few toppers.

Python: Loading the data and checking the marks column.

```
import pandas as pd

df = pd.read_csv("data/placement.csv")
df.shape      # (1000, 3)

df["placement_exam_marks"].skew()      # 0.84
df["placement_exam_marks"].describe()   # the table above
```

7. Finding the fences and the outliers

Key point: $Q_1 = 17$ and $Q_3 = 44$ give an IQR of 27 and fences of -23.5 and 84.5 ; 15 students lie above the upper fence and none below the lower one.

Figure 2 shows the box plot of the marks above their histogram. The box runs from $Q_1 = 17$ to $Q_3 = 44$, so the IQR is 27, and the fences sit at -23.5 and 84.5 (Section 4).

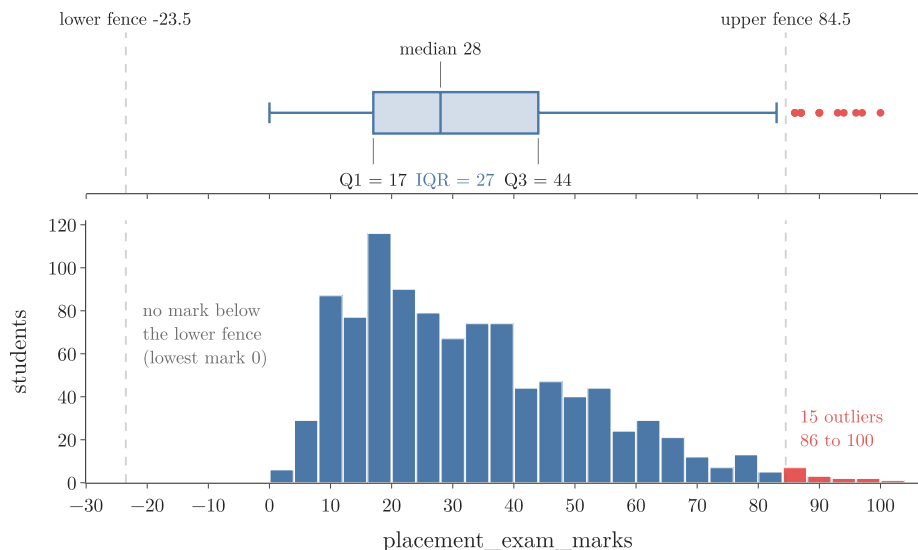


Figure 2: The placement exam marks: box plot with Q_1 , median and Q_3 , the two fences, and the 15 outliers in red

- **Upper fence, 84.5:** 15 students scored above it, with marks from 86 to 100. In the box plot they are the red dots on the right.
- **Lower fence, -23.5 :** marks cannot be negative, so no student lies below it. With a right-skewed column, all the outliers sit on the long-tail side.

The 15 outliers:

Mark	Students	Rows
86	3	40, 61, 771
87	4	283, 290, 311, 685
90	3	162, 324, 730
93, 94, 96, 97, 100	1 each	134, 9, 630, 846, 917

Python: The quartiles, the IQR and the fences.

```
marks = df["placement_exam_marks"]
```

```
percentile25 = marks.quantile(0.25)    # 17.0
```

```
percentile75 = marks.quantile(0.75)    # 44.0
```

```
iqr = percentile75 - percentile25      # 27.0
```

```
upper_limit = percentile75 + 1.5 * iqr  # 84.5
```

```
lower_limit = percentile25 - 1.5 * iqr  # -23.5
```

```
df[marks > upper_limit]    # 15 rows
```

```
df[marks < lower_limit]    # 0 rows
```

`quantile(0.25)` returns the 25th percentile; `quantile` takes the fraction (0.25), not the percent (25).

8. Trimming in code

Key point: Keeping only the rows inside the fences removes the 15 outliers and leaves 985 rows.

Trimming is a filter: keep the rows whose mark lies between the two fences. Since no mark is below the lower fence, keeping the marks below 84.5 is enough here.

Python: Trimming.

```
# keep the rows with lower <= mark <= upper
```

```
new_df = df[marks.between(lower_limit, upper_limit)]
```

```
new_df.shape    # (985, 3)
```

Figure 3 (middle row) shows the result. The histogram barely changes: only its thin right tail is gone. The box plot loses its row of red dots.

The mean drops from 32.23 to 31.34 and the skewness from 0.84 to 0.65.

8.1 Why a new outlier appears after trimming

Key point: The box plot of the trimmed data computes new fences from the 985 remaining rows, so a value that was inside before can now be outside.

The trimmed box plot in Figure 3 still shows one red dot, at a mark of 83. It is not a mistake. A box plot always computes its fences from the data it is given.

For the 985 trimmed rows, Q_3 drops from 44 to 43, so the new upper fence is $43 + 1.5 \times (43 - 17) = 82$. The mark 83, safely inside the old fence of 84.5, is now just outside the new one.

This is no reason to trim again: each round would move the fences in and flag new values. We detect once, with the fences of the original data, and treat once.

9. Capping in code

Key point: Each mark above 84.5 becomes 84.5 (and each mark below -23.5 would become -23.5); all 1,000 rows stay.

Capping goes through the column value by value, exactly as in Note 42:

- above the upper fence: replace it with the upper fence;
- below the lower fence: replace it with the lower fence;
- otherwise: leave it as it is.

NumPy's `np.where(condition, value if true, value if false)` does this for the whole column; two conditions need two calls, one inside the other.

Python: Capping with `np.where`.

```
import numpy as np
```

```
new_df_cap = df.copy()
col = new_df_cap["placement_exam_marks"]
new_df_cap["placement_exam_marks"] = np.where(
    col > upper_limit,
    upper_limit,
    np.where(col < lower_limit, lower_limit, col))
```

```
new_df_cap.shape    # (1000, 3): no row lost
```

`df.copy()` makes an independent copy, so the original `df` keeps its values for the comparison below.

The result, compared with the original and the trimmed column:

	Original	Trimmed	Capped
Rows	1,000	985	1,000
Mean	32.23	31.34	32.14
Standard deviation	19.13	17.86	18.87
Maximum	100	83	84.5
Skewness	0.84	0.65	0.76

Figure 3 (bottom row) shows the capped column. The 15 outliers now all sit at 84.5, so the

histogram rises in that one spot (the orange bar). The box plot has no dots: its whisker ends exactly at the fence.

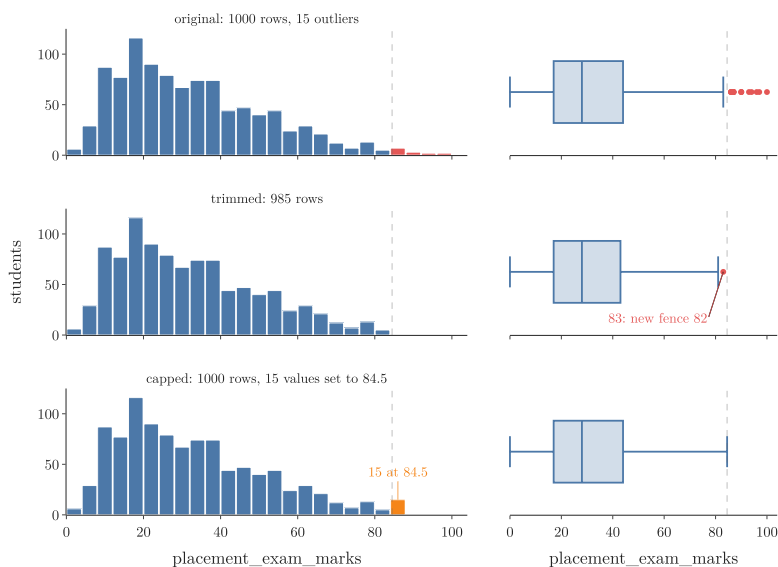


Figure 3: Before and after: the original column, the trimmed column (985 rows) and the capped column (1,000 rows, outliers moved onto 84.5); histograms on the left, box plots on the right, the upper fence dashed

Extra: pandas has a one-line shortcut for capping, `clip`. It gives exactly the same column as the two nested `np.where` calls.

```
marks.clip(lower_limit, upper_limit)
```

10. Learning the fences on the training set

Key point: The fences are learned from the data, so they should be learned on the training set only and then applied to the test set.

Extra: The steps above compute Q_1 and Q_3 on all 1,000 rows, before any train-test split. This lets the test rows influence the fences, the same data leakage that Note 24 (Section 7.2) avoids for scaling and Note 42 (Section 11) avoids for the z-score limits. The cleaner order is:

1. Split the data into training and test sets.
2. Compute Q_1 , Q_3 , the IQR and the fences on the training set only.
3. Apply the same fences to the training and the test set.

With an 80/20 split (`random_state=42`), the 800 training rows have the same $Q_1 = 17$ and $Q_3 = 44$, so the fences stay at -23.5 and 84.5 :

	Rows	Outliers (training fences)
Training set	800	14: marks 86 to 97
Test set	200	1: mark 100

Quartiles move very little between 800 and 1,000 rows, so here the result is the same. Trimming is done on the training set only (786 rows remain), since we cannot delete test rows at prediction time. Capping can be applied to both sets with the training fences; the Notebook shows both steps.

11. Strengths and limits of the IQR method

Key point: The method is simple, works on skewed columns and is not pulled by the outliers it looks for; it is only a rule of thumb.

- **Simple:** two percentiles give both fences.
- **Shape-free:** it needs no bell shape, so it fits skewed columns.
- **Robust:** the quartiles are not pulled by extreme values (Section 4, Extra).

Extra: Two things to keep in mind.

- **On a long tail, real values get flagged.** In a strongly skewed column the far tail is often genuine (a few very high incomes, a few toppers). The fences flag it all the same, so we still decide, as in Note 41 (Section 4), whether those values are errors or real.
- **One side may never be used.** For a right-skewed column with a natural lower bound, such as marks starting at 0, the lower fence is often below every possible value. That is expected, not a bug.

12. Summary

Step	What we do	On the placement marks
Check the shape	plot the column; the method is for skewed columns	right-skewed, skewness 0.84
Quartiles	$Q_1 = 25\text{th}$, $Q_3 = 75\text{th}$ percentile, $\text{IQR} = Q_3 - Q_1$	$Q_1 = 17$, $Q_3 = 44$, $\text{IQR} = 27$
Detect	fences $Q_1 - 1.5 \text{ IQR}$ and $Q_3 + 1.5 \text{ IQR}$	-23.5 and 84.5 ; 15 outliers, all above
Trim	keep the rows inside the fences	985 rows left
Cap	move values beyond a fence onto it	1,000 rows; maximum 84.5
Better practice	learn the fences on the training set	same fences; 14 outliers in train, 1 in test

	Z-score method (Note 42)	IQR method (this Note)
Column shape	roughly normal	skewed
Built on	mean and standard deviation	Q_1 and Q_3
Limits	$\mu \pm 3\sigma$	$Q_1 - 1.5 \text{ IQR}$, $Q_3 + 1.5 \text{ IQR}$
Pulled by outliers	yes	hardly

- The IQR method is for skewed columns; the z-score method of Note 42 is for normal ones.
- The IQR is $Q_3 - Q_1$, the width of a box plot's box.
- The fences are $Q_1 - 1.5 \text{ IQR}$ and $Q_3 + 1.5 \text{ IQR}$: the same limits a box plot uses for its dots.
- Values outside the fences are outliers; we trim them (drop the rows) or cap them (set them to the fence).
- A box plot of trimmed data computes new fences, so a new dot can appear; we do not trim again.
- The fences should be learned on the training set only.

13. Key terms

Term	Meaning
IQR method (IQR proximity rule)	Outlier detection that flags values beyond 1.5 IQR outside the box (Q_1 to Q_3); for skewed columns
Percentile	The value below which a given percent of the column lies

Term	Meaning
Quartile	Q_1 (25th percentile) and Q_3 (75th percentile): they cut the sorted column into quarters
Interquartile range (IQR)	$Q_3 - Q_1$: the width of the middle half of the data
Fence	$Q_1 - 1.5 \text{ IQR}$ (lower) or $Q_3 + 1.5 \text{ IQR}$ (upper); values past it are outliers
<code>quantile</code>	pandas method that returns a percentile, given as a fraction (0.25 for the 25th)